

Course 5332

5 Credits: 4

Program core

Source-grounded

# AI for Robotics

Study the official course objectives in the source syllabus.

## OFFICIAL SOURCE DECLARATION

### How this lesson was prepared

The supplied official SITTTR syllabus PDF for Course 5332 is the authoritative source for course information, objectives, course outcomes, CO-PO mapping, modules, topics, activities, assessment structure and references. Explanations, study prompts and Malayalam support are educational elaboration and are not presented as additional official syllabus text.

**Source file:** 5332.pdf from the uploaded official SITTTR syllabus archive. **Course code and title:** preserved from the source.

## COURSE DASHBOARD

### Official course information

#### Course information

| Item                 | Official information                     |
|----------------------|------------------------------------------|
| Programme            | Diploma in Automation and Robotics       |
| Course code          | 5332                                     |
| Course title         | AI for Robotics                          |
| Semester             | 5 Credits: 4                             |
| Course category      | Program core                             |
| Periods per week     | 4 (L:4 T:0 P:0) Periods per semester: 60 |
| Periods per semester | 60                                       |

| Item         | Official information                                 |
|--------------|------------------------------------------------------|
| Credits      | 4                                                    |
| Prerequisite | See the official syllabus PDF                        |
| Assessment   | Use the official assessment section reproduced below |

## Modules

4 official module sections detected.

## Topic entries

11 source-aligned topic entries retained.

## Study mode

Theory and application emphasis.

### STUDY METHOD

## How to use this lesson

### First pass

Read the official source declaration, dashboard and module transcript. Mark unfamiliar terms without changing the official terminology.

### Second pass

Use each topic card to build a definition, principle, steps or components, application and exam answer. Attempt the Q&A before opening the answers.

### Practical evidence

Where the course is laboratory or workshop based, maintain an observation notebook with procedure, instruments, readings, safety points and conclusion.

## Revision pass

Return to the source excerpt, coverage table, activities and model paper. The generated paper is practice material, not an official examination paper.

### LEARNING OUTCOMES

## Objectives, outcomes and mapping

### Course objectives

- Study the official course objectives in the source syllabus.

#### Course outcomes

| CO  | Official outcome                                                                     | Study level           |
|-----|--------------------------------------------------------------------------------------|-----------------------|
| CO1 | 14 principles and techniques Understand learning concepts and learning from Applying | See official syllabus |
| CO2 | 15 outcomes Understand the importance, applicability, and Applying                   | See official syllabus |
| CO3 | 15 strength of AI. Familiarize with programming languages in AI and Understanding    | See official syllabus |
| CO4 | 14 understand the architecture of Expert system.                                     | See official syllabus |

#### CO-PO mapping evidence

| Row | Extracted official mapping |
|-----|----------------------------|
| CO1 | CO1 2 1                    |
| CO2 | CO2 2 1                    |
| CO3 | CO3 2 1                    |
| CO4 | CO4 3 2                    |

### COVERAGE AUDIT

## Complete syllabus coverage

Every detected official module is represented below. Each module contains topic cards and an expandable official syllabus transcript to preserve source terminology.

#### Module coverage

| Module   | Official heading                                                            | Topic entries | Hours        |
|----------|-----------------------------------------------------------------------------|---------------|--------------|
| Module 1 | Familiarize students with Artificial Intelligence principles and techniques | 3             | As specified |
| Module 2 | Understand learning concepts and learning from outcomes                     | 3             | As specified |
| Module 3 | Understand the importance, applicability, and strength of AI.               | 3             | As specified |
| Module 4 | architecture of Expert system.                                              | 2             | As specified |

#### MODULE 1

## Familiarize students with Artificial Intelligence principles and techniques

This module is presented from the official syllabus scope for AI for Robotics. Use the topic cards for learning and the source transcript for exact coverage.

**As specified**  
official hours

**CO-linked**  
see outcomes

### Learning goals

- Explain the official Module 1 scope and its relationship to the course outcomes.
- Recognise the terminology, sequence, components and evidence expected in examination answers.
- Connect at least one official topic to a practical, industrial, laboratory or design context.

### Key facts

- Official module title: Familiarize students with Artificial Intelligence principles and techniques
- Official duration: As specified
- Official topic entries captured: 3

- The full source excerpt is preserved below for coverage checking.

## 4 Understanding historical background

4 Understanding historical background is an official topic in Module 1 of AI for Robotics. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

### Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

### Study points

- Define or identify 4 Understanding historical background using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

### Engineering or professional relevance

In Polytechnic practice, 4 understanding historical background should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

#### Topic study guide

| Aspect              | What to prepare                                                                   |
|---------------------|-----------------------------------------------------------------------------------|
| Definition/identity | What 4 Understanding historical background means in the official course context   |
| Study lens          | Principle → components or stages → result → application                           |
| Exam evidence       | Definition, labelled explanation, comparison, procedure or example as appropriate |

**Exam tip:** Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

**Common mistakes:** Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

**Malayalam support:** Module 1-4 Understanding historical background  
പ്രാചീനകാലത്തെ പദങ്ങൾ definition/meaning പദങ്ങൾക്കുള്ള ഉദാഹരണം. പദങ്ങൾ പദങ്ങൾ പദങ്ങൾ,  
steps, application പദങ്ങൾ പദങ്ങൾക്കുള്ള പദങ്ങൾ. Technical terms English-ൽ പദങ്ങൾ  
പദങ്ങൾക്കുള്ള പദങ്ങൾ.

## Understand various application areas of AI

Understand various application areas of AI is an official topic in Module 1 of AI for Robotics. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

### Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

### Study points

- Define or identify Understand various application areas of AI using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

### Engineering or professional relevance

In Polytechnic practice, understand various application areas of ai should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

#### Topic study guide

| Aspect              | What to prepare                                                                      |
|---------------------|--------------------------------------------------------------------------------------|
| Definition/identity | What Understand various application areas of AI means in the official course context |
| Study lens          | Principle → components or stages → result → application                              |
| Exam evidence       | Definition, labelled explanation, comparison, procedure or example as appropriate    |

**Exam tip:** Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

**Common mistakes:** Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.



**Malayalam support:** Module 1-ഈ മോഡ്യൂളിൽ വിവിധ AI പ്രയോഗ മേഖലകളെക്കുറിച്ച് മനസ്സിലാക്കുക. നിർവ്വചനം/അർത്ഥം, ഘട്ടങ്ങൾ, പ്രയോഗ മേഖലകൾ എന്നിവയെക്കുറിച്ച്. Technical terms English-ൽ നിന്ന് മലയാളത്തിലേക്ക് പരിഭാഷിക്കുക.

## **Understand about the basics of expert systems 4 Understanding Artificial intelligence - Introduction, its importance, The Turing test, Foundations of artificial intelligence, A brief historical overview. Application areas of AI - vision and speech processing, robotics, expert systems -basic overview**

Understand about the basics of expert systems 4 Understanding Artificial intelligence - Introduction, its importance, The Turing test, Foundations of artificial intelligence, A brief historical overview. Application areas of AI - vision and speech processing, robotics, expert systems -basic overview is an official topic in Module 1 of AI for Robotics. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

### **Concept and explanation**

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

### **Study points**

- Define or identify Understand about the basics of expert systems 4 Understanding Artificial intelligence - Introduction, its importance, The Turing test, Foundations of artificial intelligence, A brief historical overview. Application areas of AI - vision and speech processing, robotics, expert systems -basic overview using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

### **Engineering or professional relevance**

In Polytechnic practice, understand about the basics of expert systems 4 understanding artificial intelligence - introduction, its importance, the turing test, foundations of artificial intelligence, a brief historical overview. application areas of ai - vision and speech processing, robotics, expert systems -basic overview should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

#### **Topic study guide**

| Aspect | What to prepare |
|--------|-----------------|
|--------|-----------------|

| Aspect              | What to prepare                                                                                                                                                                                                                                                                                                                                        |
|---------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Definition/identity | What Understand about the basics of expert systems 4 Understanding Artificial intelligence - Introduction, its importance, The Turing test, Foundations of artificial intelligence, A brief historical overview. Application areas of AI - vision and speech processing, robotics, expert systems -basic overview means in the official course context |
| Study lens          | Principle → components or stages → result → application                                                                                                                                                                                                                                                                                                |
| Exam evidence       | Definition, labelled explanation, comparison, procedure or example as appropriate                                                                                                                                                                                                                                                                      |

**Exam tip:** Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

**Common mistakes:** Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

**Malayalam support:** Module 1- Understand about the basics of expert systems 4 Understanding Artificial intelligence - Introduction, its importance, The Turing test, Foundations of artificial intelligence, A brief historical overview. Application areas of AI - vision and speech processing, robotics, expert systems -basic overview definition/meaning . . . . ., steps, application . . . . . Technical terms English- . . . . .

## Official syllabus detail and terminology

The following excerpt preserves official syllabus terminology for this module.  
Educational explanations below are separate elaboration.

Familiarize students with Artificial Intelligence principles and techniques

|                                        |                                               |   |
|----------------------------------------|-----------------------------------------------|---|
| Familiarize with basic AI concepts and |                                               |   |
| M1.01                                  |                                               | 4 |
| Understanding                          | historical background                         |   |
| M1.02                                  | Understand various application areas of AI    | 6 |
| Understanding                          |                                               |   |
| M1.03                                  | Understand about the basics of expert systems | 4 |
| Understanding                          |                                               |   |

Artificial intelligence - Introduction, its importance, The Turing test, Foundations of artificial intelligence, A brief historical overview. Application areas of AI - vision and speech processing, robotics, expert systems -basic overview

**Module study routine:** Read the source wording, make a one-page concept map, practise one short answer and one structured answer, then review the Malayalam support notes.

### MODULE 2

## Understand learning concepts and learning from outcomes

This module is presented from the official syllabus scope for AI for Robotics. Use the topic cards for learning and the source transcript for exact coverage.

|                                |                           |
|--------------------------------|---------------------------|
| As specified<br>official hours | CO-linked<br>see outcomes |
|--------------------------------|---------------------------|

## Learning goals

- Explain the official Module 2 scope and its relationship to the course outcomes.
- Recognise the terminology, sequence, components and evidence expected in examination answers.
- Connect at least one official topic to a practical, industrial, laboratory or design context.

### **Key facts**

- Official module title: Understand learning concepts and learning from outcomes
- Official duration: As specified
- Official topic entries captured: 3
- The full source excerpt is preserved below for coverage checking.

## Understand the basic types of learning

Understand the basic types of learning is an official topic in Module 2 of AI for Robotics. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

### Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

### Study points

- Define or identify Understand the basic types of learning using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

### Engineering or professional relevance

In Polytechnic practice, understand the basic types of learning should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

#### Topic study guide

| Aspect              | What to prepare                                                                   |
|---------------------|-----------------------------------------------------------------------------------|
| Definition/identity | What Understand the basic types of learning means in the official course context  |
| Study lens          | Principle → components or stages → result → application                           |
| Exam evidence       | Definition, labelled explanation, comparison, procedure or example as appropriate |

**Exam tip:** Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

**Common mistakes:** Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

**Malayalam support:** Module 2-ഓരോ പഠനത്തിന്റെയും അടിസ്ഥാനപരമായ തരങ്ങൾ മനസ്സിലാക്കുക. പഠനത്തിന്റെ നിർവ്വചനം/അർത്ഥം, ഘട്ടങ്ങൾ, പ്രയോഗം എന്നിവയെക്കുറിച്ചും മനസ്സിലാക്കുക. Technical terms English-ൽ പരിഭാഷിക്കുക.

## Discuss the details of reinforcement learning 3 Understanding Familiarize with programming languages used

Discuss the details of reinforcement learning 3 Understanding Familiarize with programming languages used is an official topic in Module 2 of AI for Robotics. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

### Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

### Study points

- Define or identify Discuss the details of reinforcement learning 3 Understanding Familiarize with programming languages used using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

### Engineering or professional relevance

In Polytechnic practice, discuss the details of reinforcement learning 3 understanding familiarize with programming languages used should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

#### Topic study guide

| Aspect              | What to prepare                                                                                                                                     |
|---------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------|
| Definition/identity | What Discuss the details of reinforcement learning 3 Understanding Familiarize with programming languages used means in the official course context |
| Study lens          | Principle → components or stages → result → application                                                                                             |
| Exam evidence       | Definition, labelled explanation, comparison, procedure or example as appropriate                                                                   |

**Exam tip:** Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.



**Common mistakes:** Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

**Malayalam support:** Module 2- Discuss the details of reinforcement learning 3 Understanding Familiarize with programming languages used   
 definition/meaning . . , steps, application   
 . Technical terms English- .

## **6 Applying in AI Learning - Forms of learning, Supervised Learning Algorithms, Unsupervised Learning Algorithms, Reinforcement based learning - overview with basic elements agent, environment, action, state, reward only; Programming Language: Introduction to programming Language, LISP and PROLOG.**

6 Applying in AI Learning - Forms of learning, Supervised Learning Algorithms, Unsupervised Learning Algorithms, Reinforcement based learning - overview with basic elements agent, environment, action, state, reward only; Programming Language: Introduction to programming Language, LISP and PROLOG. is an official topic in Module 2 of AI for Robotics. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

### **Concept and explanation**

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

### **Study points**

- Define or identify 6 Applying in AI Learning - Forms of learning, Supervised Learning Algorithms, Unsupervised Learning Algorithms, Reinforcement based learning - overview with basic elements agent, environment, action, state, reward only; Programming Language: Introduction to programming Language, LISP and PROLOG. using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

### **Engineering or professional relevance**

In Polytechnic practice, 6 applying in ai learning - forms of learning, supervised learning algorithms, unsupervised learning algorithms, reinforcement based learning - overview with basic elements agent, environment, action, state, reward only; programming language: introduction to programming language, lisp and prolog. should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

| Aspect              | What to prepare                                                                                                                                                                                                                                                                                                                                     |
|---------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Definition/identity | What 6 Applying in AI Learning - Forms of learning, Supervised Learning Algorithms, Unsupervised Learning Algorithms, Reinforcement based learning - overview with basic elements agent, environment, action, state, reward only; Programming Language: Introduction to programming Language, LISP and PROLOG. means in the official course context |
| Study lens          | Principle → components or stages → result → application                                                                                                                                                                                                                                                                                             |
| Exam evidence       | Definition, labelled explanation, comparison, procedure or example as appropriate                                                                                                                                                                                                                                                                   |

**Exam tip:** Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

**Common mistakes:** Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

**Malayalam support:** Module 2- 6 Applying in AI Learning - Forms of learning, Supervised Learning Algorithms, Unsupervised Learning Algorithms, Reinforcement based learning - overview with basic elements agent, environment, action, state, reward only; Programming Language: Introduction to programming Language, LISP and PROLOG. definition/meaning . steps, application . Technical terms English- .

## Official syllabus detail and terminology

The following excerpt preserves official syllabus terminology for this module.  
Educational explanations below are separate elaboration.

Understand learning concepts and learning from outcomes

|               |                                        |   |
|---------------|----------------------------------------|---|
| M2.01         | Understand the basic types of learning | 6 |
| Understanding |                                        |   |

|               |                                               |   |
|---------------|-----------------------------------------------|---|
| M2.02         | Discuss the details of reinforcement learning | 3 |
| Understanding |                                               |   |

|          |                                             |   |
|----------|---------------------------------------------|---|
|          | Familiarize with programming languages used |   |
| M2.03    |                                             | 6 |
| Applying |                                             |   |
|          | in AI                                       |   |

|  |                 |   |
|--|-----------------|---|
|  | Series Test – I | 1 |
|--|-----------------|---|

Learning - Forms of learning, Supervised Learning Algorithms, Unsupervised Learning Algorithms, Reinforcement based learning - overview with basic elements agent, environment, action, state, reward only; Programming Language: Introduction to programming Language, LISP and PROLOG.

**Module study routine:** Read the source wording, make a one-page concept map, practise one short answer and one structured answer, then review the Malayalam support notes.

### MODULE 3

## Understand the importance, applicability, and strength of AI.

This module is presented from the official syllabus scope for AI for Robotics. Use the topic cards for learning and the source transcript for exact coverage.

**As specified**  
official hours

**CO-linked**  
see outcomes

## Learning goals

- Explain the official Module 3 scope and its relationship to the course outcomes.
- Recognise the terminology, sequence, components and evidence expected in examination answers.
- Connect at least one official topic to a practical, industrial, laboratory or design context.

## Key facts

- Official module title: Understand the importance, applicability, and strength of AI.
- Official duration: As specified
- Official topic entries captured: 3
- The full source excerpt is preserved below for coverage checking.

## 5 Understanding expert systems Understand the knowledge representation

5 Understanding expert systems Understand the knowledge representation is an official topic in Module 3 of AI for Robotics. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

### Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

### Study points

- Define or identify 5 Understanding expert systems Understand the knowledge representation using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

### Engineering or professional relevance

In Polytechnic practice, 5 understanding expert systems understand the knowledge representation should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

#### Topic study guide

| Aspect              | What to prepare                                                                                                  |
|---------------------|------------------------------------------------------------------------------------------------------------------|
| Definition/identity | What 5 Understanding expert systems Understand the knowledge representation means in the official course context |
| Study lens          | Principle → components or stages → result → application                                                          |
| Exam evidence       | Definition, labelled explanation, comparison, procedure or example as appropriate                                |

**Exam tip:** Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

**Common mistakes:** Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related

but different term.

**Malayalam support:** Module 3-5 Understanding expert systems Understand the knowledge representation ധാരണാപ്രകാരം definition/meaning പ്രകാരം. ധാരണാ പ്രകാരം, steps, application പ്രകാരം. Technical terms English-ഈ പ്രകാരം പ്രകാരം.

## 4 Understanding schemes in expert systems

4 Understanding schemes in expert systems is an official topic in Module 3 of AI for Robotics. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

### Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

### Study points

- Define or identify 4 Understanding schemes in expert systems using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

### Engineering or professional relevance

In Polytechnic practice, 4 understanding schemes in expert systems should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

#### Topic study guide

| Aspect              | What to prepare                                                                     |
|---------------------|-------------------------------------------------------------------------------------|
| Definition/identity | What 4 Understanding schemes in expert systems means in the official course context |
| Study lens          | Principle → components or stages → result → application                             |
| Exam evidence       | Definition, labelled explanation, comparison, procedure or example as appropriate   |

**Exam tip:** Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

**Common mistakes:** Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.



**Malayalam support:** Module 3-4 Understanding schemes in expert systems  
പ്രദേശീകരണത്തിന് പ്രദേശീകരണ definition/meaning പ്രദേശീകരണത്തിന്. പ്രദേശീകരണ പ്രദേശീകരണ, steps, application പ്രദേശീകരണ പ്രദേശീകരണ പ്രദേശീകരണ. Technical terms English- പ്രദേശീകരണ പ്രദേശീകരണ.

**Familiarize with expert system tools 6 EXPERT SYSTEMS- Definition - Features of an expert system - Organization - Characteristics - Prospector - Knowledge Representation in expert systems - Expert system tools - MYCIN - EMYCIN. Familiarize with programming languages in AI and understand the**

Familiarize with expert system tools 6 EXPERT SYSTEMS- Definition - Features of an expert system - Organization - Characteristics - Prospector - Knowledge Representation in expert systems - Expert system tools - MYCIN - EMYCIN. Familiarize with programming languages in AI and understand the is an official topic in Module 3 of AI for Robotics. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

### Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

### Study points

- Define or identify Familiarize with expert system tools 6 EXPERT SYSTEMS- Definition - Features of an expert system - Organization - Characteristics - Prospector - Knowledge Representation in expert systems - Expert system tools - MYCIN - EMYCIN. Familiarize with programming languages in AI and understand the using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

### Engineering or professional relevance

In Polytechnic practice, familiarize with expert system tools 6 expert systems- definition - features of an expert system - organization - characteristics - prospector - knowledge representation in expert systems - expert system tools - mycin - emycin. familiarize with programming languages in ai and understand the should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

#### Topic study guide

| Aspect | What to prepare |
|--------|-----------------|
|--------|-----------------|

| Aspect              | What to prepare                                                                                                                                                                                                                                                                                                                               |
|---------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Definition/identity | What Familiarize with expert system tools 6 EXPERT SYSTEMS- Definition – Features of an expert system – Organization – Characteristics – Prospector – Knowledge Representation in expert systems – Expert system tools – MYCIN – EMYCIN. Familiarize with programming languages in AI and understand the means in the official course context |
| Study lens          | Principle → components or stages → result → application                                                                                                                                                                                                                                                                                       |
| Exam evidence       | Definition, labelled explanation, comparison, procedure or example as appropriate                                                                                                                                                                                                                                                             |

**Exam tip:** Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

**Common mistakes:** Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

**Malayalam support:** Module 3-എന്നുപരിചയപ്പെടുത്തുക 6 EXPERT SYSTEMS- Definition – Features of an expert system – Organization – Characteristics – Prospector – Knowledge Representation in expert systems – Expert system tools – MYCIN – EMYCIN. Familiarize with programming languages in AI and understand the ഏകദേശം definition/meaning എങ്ങനെ. എങ്ങനെ എങ്ങനെ, steps, application എങ്ങനെ എങ്ങനെ എങ്ങനെ. Technical terms English-എന്നുപരിചയപ്പെടുത്തുക.

## Official syllabus detail and terminology

The following excerpt preserves official syllabus terminology for this module.  
Educational explanations below are separate elaboration.

Understand the importance, applicability, and strength of AI.

|                                                |                                      |                |
|------------------------------------------------|--------------------------------------|----------------|
| Discuss different features and organization of |                                      |                |
| M3.01                                          |                                      | 5              |
| Understanding                                  | expert systems                       |                |
|                                                | Understand the knowledge             | representation |
| M3.02                                          |                                      | 4              |
| Understanding                                  | schemes in expert systems            |                |
| M3.03                                          | Familiarize with expert system tools | 6              |

EXPERT SYSTEMS- Definition – Features of an expert system – Organization – Characteristics – Prospector – Knowledge Representation in expert systems – Expert system tools – MYCIN – EMYCIN.

Familiarize with programming languages in AI and understand the

**Module study routine:** Read the source wording, make a one-page concept map, practise one short answer and one structured answer, then review the Malayalam support notes.

### MODULE 4

## architecture of Expert system.

This module is presented from the official syllabus scope for AI for Robotics. Use the topic cards for learning and the source transcript for exact coverage.

**As specified**  
official hours

**CO-linked**  
see outcomes

### Learning goals

- Explain the official Module 4 scope and its relationship to the course outcomes.

- Recognise the terminology, sequence, components and evidence expected in examination answers.
- Connect at least one official topic to a practical, industrial, laboratory or design context.

### **Key facts**

- Official module title: architecture of Expert system.
- Official duration: As specified
- Official topic entries captured: 2
- The full source excerpt is preserved below for coverage checking.

## 10 robotics

10 robotics is an official topic in Module 4 of AI for Robotics. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

### Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

### Study points

- Define or identify 10 robotics using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

### Engineering or professional relevance

In Polytechnic practice, 10 robotics should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

#### Topic study guide

| Aspect              | What to prepare                                                                   |
|---------------------|-----------------------------------------------------------------------------------|
| Definition/identity | What 10 robotics means in the official course context                             |
| Study lens          | Principle → components or stages → result → application                           |
| Exam evidence       | Definition, labelled explanation, comparison, procedure or example as appropriate |

**Exam tip:** Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

**Common mistakes:** Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

**Malayalam support:** Module 4-10 robotics റോബോട്ടിക്സ് റോബോട്ട്  
definition/meaning റോബോട്ടിക്സ്. റോബോട്ട് റോബോട്ടിക്സ്, steps, application  
റോബോട്ട് റോബോട്ടിക്സ് റോബോട്ടിക്സ്. Technical terms English-റോബോട്ട് റോബോട്ടിക്സ്.

**Discuss the ethics and risks of AI usage in robotics 4 AI In Robotics and Applications- Robotic perception, localization, mapping- configuring space, planning uncertain movements, dynamics and control of movement, Ethics and risks of artificial intelligence in robotics Text / Reference T/R Book Title/Author Stuart Russel, Peter Norvig, “Artificial Intelligence: A modern approach” Pearson T1 Education, India,2016 Nilsson, Nils J.“Principles of Artificial Intelligence” Springer Science & Business T2 Media. Deepak Khemani, A First Course in Artificial Intelligence, McGraw Hill Education (India) R1 2013 N.P.Padhy: Artificial Intelligence and Intelligent Systems, Oxford University Press, R2 2009. Negnevitsky, M, “Artificial Intelligence: A guide to Intelligence Systems”, R3 Harlow:AddisonWesley,2002**

Discuss the ethics and risks of AI usage in robotics 4 AI In Robotics and Applications- Robotic perception, localization, mapping- configuring space, planning uncertain movements, dynamics and control of movement, Ethics and risks of artificial intelligence in robotics Text / Reference T/R Book Title/Author Stuart Russel, Peter Norvig, “Artificial Intelligence: A modern approach” Pearson T1 Education, India,2016 Nilsson, Nils J.“Principles of Artificial Intelligence” Springer Science & Business T2 Media. Deepak Khemani, A First Course in Artificial Intelligence, McGraw Hill Education (India) R1 2013 N.P.Padhy: Artificial Intelligence and Intelligent Systems, Oxford University Press, R2 2009. Negnevitsky, M, “Artificial Intelligence: A guide to Intelligence Systems”, R3 Harlow:AddisonWesley,2002 is an official topic in Module 4 of AI for Robotics. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

### **Concept and explanation**

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

### **Study points**

- Define or identify Discuss the ethics and risks of AI usage in robotics 4 AI In Robotics and Applications- Robotic perception, localization, mapping- configuring space, planning uncertain movements, dynamics and control of movement, Ethics and risks of artificial intelligence in robotics Text / Reference T/R Book Title/Author Stuart Russel, Peter Norvig, “Artificial Intelligence: A modern approach” Pearson T1 Education, India,2016 Nilsson, Nils J.“Principles of Artificial Intelligence” Springer Science & Business T2 Media. Deepak Khemani, A First Course in Artificial Intelligence, McGraw Hill Education (India) R1 2013 N.P.Padhy: Artificial Intelligence



and Intelligent Systems, Oxford University Press, R2 2009. Negnevitsky, M, "Artificial Intelligence: A guide to Intelligence Systems", R3 Harlow:AddisonWesley,2002 using standard technical terminology.

- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

## Engineering or professional relevance

In Polytechnic practice, discuss the ethics and risks of ai usage in robotics 4 ai in robotics and applications- robotic perception, localization, mapping- configuring space, planning uncertain movements, dynamics and control of movement, ethics and risks of artificial intelligence in robotics text / reference t/r book title/author stuart russel, peter norvig, "artificial intelligence: a modern approach" pearson t1 education, india,2016 nilsson, nils j."principles of artificial intelligence" springer science & business t2 media. deepak khemani, a first course in artificial intelligence, mcgraw hill education (india) r1 2013 n.p.padhy: artificial intelligence and intelligent systems, oxford university press, r2 2009. negnevitsky, m, "artificial intelligence: a guide to intelligence systems", r3 harlow:addisonwesley,2002 should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

### Topic study guide

| Aspect              | What to prepare                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
|---------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Definition/identity | What Discuss the ethics and risks of AI usage in robotics 4 AI In Robotics and Applications- Robotic perception, localization, mapping- configuring space, planning uncertain movements, dynamics and control of movement, Ethics and risks of artificial intelligence in robotics Text / Reference T/R Book Title/Author Stuart Russel, Peter Norvig, "Artificial Intelligence: A modern approach" Pearson T1 Education, India,2016 Nilsson, Nils J."Principles of Artificial Intelligence" Springer Science & Business T2 Media. Deepak Khemani, A First Course in Artificial Intelligence, McGraw Hill Education (India) R1 2013 N.P.Padhy: Artificial Intelligence and Intelligent Systems, Oxford University Press, R2 2009. Negnevitsky, M, "Artificial Intelligence: A guide to Intelligence Systems", R3 Harlow:AddisonWesley,2002 means in the official course context |
| Study lens          | Principle → components or stages → result → application                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| Exam evidence       | Definition, labelled explanation, comparison, procedure or example as appropriate                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |

**Exam tip:** Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

**Common mistakes:** Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

**Malayalam support:** Module 4- Discuss the ethics and risks of AI usage in robotics 4 AI In Robotics and Applications- Robotic perception, localization, mapping- configuring space, planning uncertain movements, dynamics and control of movement, Ethics and risks of artificial intelligence in robotics Text / Reference T/R Book Title/Author Stuart Russel, Peter Norvig, "Artificial Intelligence: A modern approach" Pearson T1 Education, India,2016 Nilsson, Nils J."Principles of Artificial Intelligence" Springer Science & Business T2 Media. Deepak Khemani, A First Course in Artificial Intelligence, McGraw Hill Education (India) R1 2013 N.P.Padhy: Artificial Intelligence and Intelligent Systems, Oxford University Press, R2 2009. Negnevitsky, M, "Artificial Intelligence: A guide to Intelligence Systems", R3 Harlow:AddisonWesley,2002 definition/meaning steps, application Technical terms English-

## Official syllabus detail and terminology

The following excerpt preserves official syllabus terminology for this module.  
Educational explanations below are separate elaboration.

architecture of Expert system.

|       |                                                         |    |
|-------|---------------------------------------------------------|----|
| M4.01 | Familiarize with AI applications pertaining to robotics | 10 |
| M4.02 | Discuss the ethics and risks of AI usage in robotics    | 4  |

Series Test – II 1

AI In Robotics and Applications- Robotic perception, localization, mapping-  
configuring  
space, planning uncertain movements, dynamics and control of movement, Ethics  
and risks  
of artificial intelligence in robotics

Text / Reference

| T/R              | Book Title/Author                                                                                                                                                                                            |
|------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Pearson<br>T1    | Stuart Russel, Peter Norvig, "Artificial Intelligence: A modern approach"                                                                                                                                    |
| & Business<br>T2 | Education, India, 2016<br>Nilsson, Nils J. "Principles of Artificial Intelligence" Springer Science<br>Media.<br>Deepak Khemani, A First Course in Artificial Intelligence, McGraw Hill<br>Education (India) |

**Module study routine:** Read the source wording, make a one-page concept map, practise one short answer and one structured answer, then review the Malayalam support notes.

## RAPID REFERENCE

## Concept and formula bank

### Answer structure

Definition → Principle → Components/steps → Application → Conclusion

Use this sequence for descriptive questions when the topic does not require a numerical formula.

## Practical record

Aim → Apparatus → Procedure → Observation → Result → Safety

Use this sequence for laboratory, workshop and field activities.

### RAPID REVISION

## Diagram and source-transcript library

### Module 1 source and topic cards

Jump to official terminology, topic explanations and study guidance.

### Module 2 source and topic cards

Jump to official terminology, topic explanations and study guidance.

### Module 3 source and topic cards

Jump to official terminology, topic explanations and study guidance.

### Module 4 source and topic cards

Jump to official terminology, topic explanations and study guidance.

### OFFICIAL SELF-LEARNING

# Activities from the syllabus

Use the extracted official activity wording as the record of required self-learning work.

The official PDF does not expose a separate activities heading in the extracted text; consult the source PDF pages for the exact activity list.

## EXAMINATION PREPARATION

### Assessment methodology

The following source extract preserves the assessment information detected in the official PDF. Verify mark conversions and institutional updates against the current source before an examination.

The official assessment structure is reproduced from the supplied PDF.

**Practice-paper notice:** The model paper below is generated for revision and is not an official examination paper.

## ACTIVE RECALL

### Module-wise questions and answers

These are practice questions based on official topic coverage, not official examination questions.

#### Module 1

**Define or explain 4 Understanding historical background. (3 marks)**

**Answer:** Begin with the standard definition or identity of *4 Understanding historical background*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

**Malayalam support:** □ □□□□□□□□□ □□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□□□ □□□□□□.

**Define or explain Understand various application areas of AI. (3 marks)**

**Answer:** Begin with the standard definition or identity of *Understand various application areas of AI*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

**Malayalam support:** □ □□□□□□□□□ □□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□□□ □□□□□□.

**Define or explain Understand about the basics of expert systems 4 Understanding Artificial intelligence - Introduction, its importance, The Turing test, Foundations of artificial intelligence, A brief historical overview. Application areas of AI - vision and speech processing, robotics, expert systems -basic overview. (3 marks)**

**Answer:** Begin with the standard definition or identity of *Understand about the basics of expert systems 4 Understanding Artificial intelligence - Introduction, its importance, The Turing test, Foundations of artificial intelligence, A brief historical overview. Application areas of AI - vision and speech processing, robotics, expert systems -basic overview*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

**Malayalam support:** □ □□□□□□□□□ □□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□□□ □□□□□□.

## Module 2

**Define or explain Understand the basic types of learning. (3 marks)**

**Answer:** Begin with the standard definition or identity of *Understand the basic types of learning*. State the governing principle, list the key components or stages, and close

with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

**Malayalam support:** □ □□□□□□□□□ □□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□□ □□□□□□.

**Define or explain Discuss the details of reinforcement learning 3 Understanding Familiarize with programming languages used. (3 marks)**

**Answer:** Begin with the standard definition or identity of *Discuss the details of reinforcement learning 3 Understanding Familiarize with programming languages used*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

**Malayalam support:** □ □□□□□□□□□ □□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□□ □□□□□□.

**Define or explain 6 Applying in AI Learning - Forms of learning, Supervised Learning Algorithms, Unsupervised Learning Algorithms, Reinforcement based learning - overview with basic elements agent, environment, action, state, reward only; Programming Language: Introduction to programming Language, LISP and PROLOG.. (3 marks)**

**Answer:** Begin with the standard definition or identity of *6 Applying in AI Learning - Forms of learning, Supervised Learning Algorithms, Unsupervised Learning Algorithms, Reinforcement based learning - overview with basic elements agent, environment, action, state, reward only; Programming Language: Introduction to programming Language, LISP and PROLOG..* State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

**Malayalam support:** □ □□□□□□□□□ □□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□□ □□□□□□.

**Module 3**

**Define or explain 5 Understanding expert systems Understand the knowledge representation. (3 marks)**

**Answer:** Begin with the standard definition or identity of *5 Understanding expert systems Understand the knowledge representation*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

**Malayalam support:** □ □□□□□□□□□ □□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□ □□□□□□.

**Define or explain 4 Understanding schemes in expert systems. (3 marks)**

**Answer:** Begin with the standard definition or identity of *4 Understanding schemes in expert systems*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

**Malayalam support:** □ □□□□□□□□□ □□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□ □□□□□□.

**Define or explain Familiarize with expert system tools 6 EXPERT SYSTEMS- Definition - Features of an expert system - Organization - Characteristics - Prospector - Knowledge Representation in expert systems - Expert system tools - MYCIN - EMYCIN. Familiarize with programming languages in AI and understand the. (3 marks)**

**Answer:** Begin with the standard definition or identity of *Familiarize with expert system tools 6 EXPERT SYSTEMS- Definition - Features of an expert system - Organization - Characteristics - Prospector - Knowledge Representation in expert systems - Expert system tools - MYCIN - EMYCIN. Familiarize with programming languages in AI and understand the*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

**Malayalam support:** □ □□□□□□□□□ □□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□ □□□□□□.



## Module 4

### Define or explain 10 robotics. (3 marks)

**Answer:** Begin with the standard definition or identity of *10 robotics*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

**Malayalam support:** □ □□□□□□□□□ □□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□ □□□□□□.

**Define or explain Discuss the ethics and risks of AI usage in robotics 4 AI In Robotics and Applications- Robotic perception, localization, mapping- configuring space, planning uncertain movements, dynamics and control of movement, Ethics and risks of artificial intelligence in robotics Text / Reference T/R Book Title/Author Stuart Russel, Peter Norvig, "Artificial Intelligence: A modern approach" Pearson T1 Education, India,2016 Nilsson, Nils J."Principles of Artificial Intelligence" Springer Science & Business T2 Media. Deepak Khemani, A First Course in Artificial Intelligence, McGraw Hill Education (India) R1 2013 N.P.Padhy: Artificial Intelligence and Intelligent Systems, Oxford University Press, R2 2009. Negnevitsky, M, "Artificial Intelligence: A guide to Intelligence Systems", R3 Harlow:AddisonWesley,2002. (3 marks)**

**Answer:** Begin with the standard definition or identity of *Discuss the ethics and risks of AI usage in robotics 4 AI In Robotics and Applications- Robotic perception, localization, mapping- configuring space, planning uncertain movements, dynamics and control of movement, Ethics and risks of artificial intelligence in robotics Text / Reference T/R Book Title/Author Stuart Russel, Peter Norvig, "Artificial Intelligence: A modern approach" Pearson T1 Education, India,2016 Nilsson, Nils J."Principles of Artificial Intelligence" Springer Science & Business T2 Media. Deepak Khemani, A First Course in Artificial Intelligence, McGraw Hill Education (India) R1 2013 N.P.Padhy: Artificial Intelligence and Intelligent Systems, Oxford University Press, R2 2009. Negnevitsky, M, "Artificial Intelligence: A guide to Intelligence Systems", R3 Harlow:AddisonWesley,2002*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

**Malayalam support:** □ □□□□□□□□□ □□□□ definition, □□□□□□ principle/steps,

## PRACTICE ONLY

# Practice Model Paper — Not an Official Examination Paper

Attempt one short definition, one structured explanation and one long answer from every module. Use the official assessment pattern reproduced above when selecting time and marks.

## Module 1

1-mark definition: State the meaning of **4 Understanding historical background**.

3-mark explanation: Explain one principle, process or comparison from this module.

7-mark answer: Write a structured answer using a labelled diagram, table, procedure, example or application.

## Module 2

1-mark definition: State the meaning of **Understand the basic types of learning**.

3-mark explanation: Explain one principle, process or comparison from this module.

7-mark answer: Write a structured answer using a labelled diagram, table, procedure, example or application.

## Module 3

1-mark definition: State the meaning of **5 Understanding expert systems**  
**Understand the knowledge representation**.

3-mark explanation: Explain one principle, process or comparison from this module.

7-mark answer: Write a structured answer using a labelled diagram, table, procedure, example or application.

## Module 4

1-mark definition: State the meaning of **10 robotics**.

3-mark explanation: Explain one principle, process or comparison from this module.

7-mark answer: Write a structured answer using a labelled diagram, table, procedure, example or application.

### QUICK REVISION

## Exam checklist

### Before writing

- Read the command word: define, explain, compare, draw, calculate, analyse or design.
- Use the exact course terminology from the source transcript.
- Plan labels, units, sequence and marks before writing.

### Before submitting

- Check every module has been revised.
- Check diagrams, tables, formula symbols and units.
- Separate official syllabus information from personal elaboration.

### REFERENCE VOCABULARY

## Glossary

### Study glossary

| Term | Meaning in this lesson |
|------|------------------------|
|------|------------------------|

| Term              | Meaning in this lesson                                                   |
|-------------------|--------------------------------------------------------------------------|
| Course Outcome    | A capability the student should demonstrate after completing the course. |
| Module            | An official syllabus unit with defined topics and hours.                 |
| CIE               | Continuous Internal Evaluation.                                          |
| ESE               | End Semester Examination.                                                |
| Source transcript | A preserved extract of the official syllabus used for coverage checking. |

## SOURCE-SUPPORTED REFERENCES

### References and web resources

#### References listed in the official PDF

References are included in the official source PDF.

#### Web resources listed in the official PDF

No separate web-resource heading was detected in the extracted text.

## IN-PAGE SEARCH

### Search this lesson

Prepared from the supplied official SITTTTR syllabus PDF for Course 5332. Verify institutional updates against the current official curriculum.